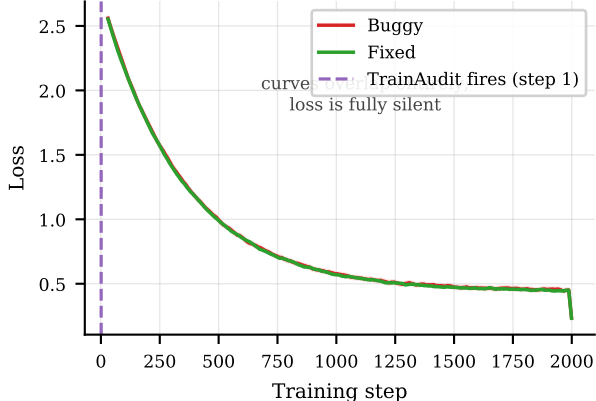
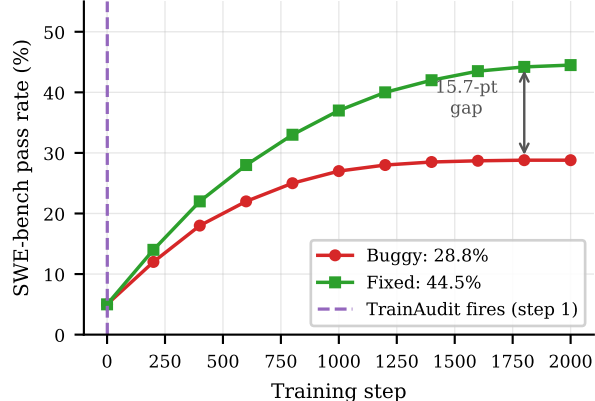


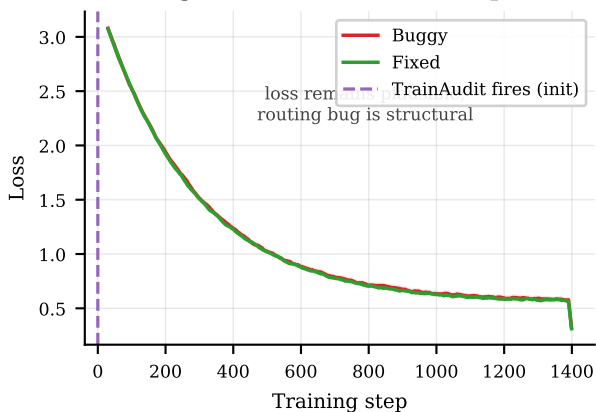
Case 1 - SFT prompt-format: training loss



Case 1 - downstream quality diverges



Case 2 - Megatron Muon+PP+MTP: pretrain loss



Case 2 - verifier output @ init

snapshot: tied_group_id = word_embeddings

PP rank	embed tag	optimizer
rank 0	yes	AdamW
rank N (MTP-only)	no	Muon

Expected: one tied group → one optimizer rule
Observed: {AdamW, Muon}

FIRE before first optimizer step